# Decision SincNet: Neurocognitive models of decision making that predict cognitive processes from neural signals

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# Sections

1 Neurocognitve Modeling and Decision Making

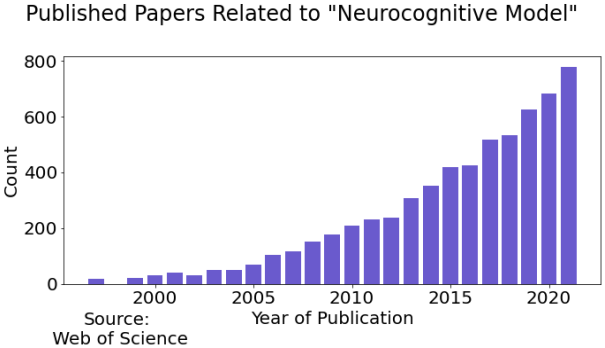
2 Decision SincNet: Bridging Neural Network and Drift-Diffusion Model

3 Model Evaluation and Results

# Neurocognitve Modeling and Decision Making

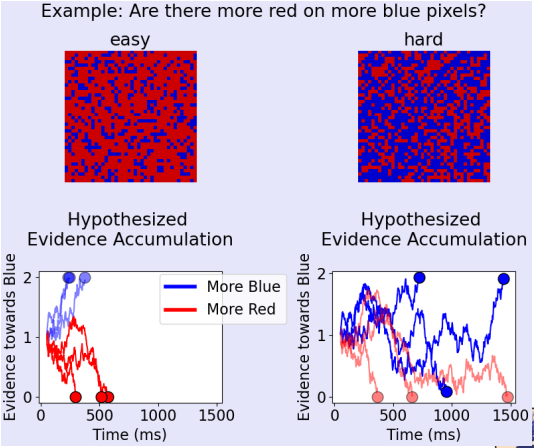
# Neurocognitive Modeling: what, why, and how

- Cognitive models are formal mathematical models that aim to describe the processes (e.g., memory, information processing) that give rise to observable behaviors.
- Neurocognitive modeling has a growing interest in the last two decades to link neural activities and cognitive processes
- One of the approaches is directly using neural data as an input to predict parameters that describe cognitive processes.

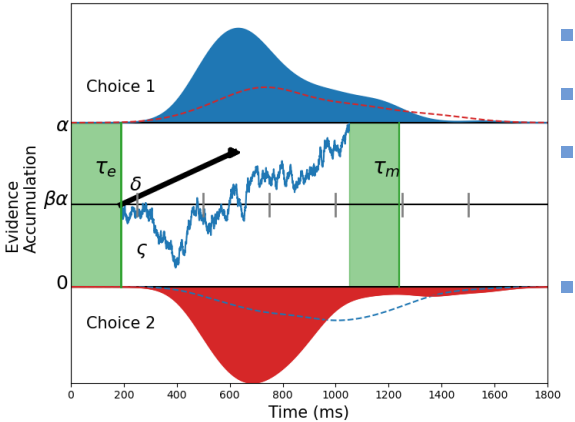


# Decision Making and Sequential Sampling

- Everyday we rapidly translate sensory information in behavioral responses
- Behavioral Patterns can be consistently observed: People respond faster when choices are easier, and people respond slower when choices are difficult.
- Sequential Sampling models suggest we continuously sample and integrate noisy evidence until a threshold is reached, like a Random Walk Process.
- Patterns of accumulation to a common threshold have be shown in neural recordings.



# Drift-Diffusion Model




- Drift -diffusion model can be used to describe Choice Response Time in Perceptual Decision Making.
- The model divides the time to respond into three intervals – perceptual encoding, decision making, and motor execution.
- The decision-process is evidence accumulation to a boundary which triggers the motor response. This process is parameterized as boundary separation $\alpha$, starting point of the random walk $\beta$, evidence accumulation or drift rate $\delta$, the variability in evidence accumulation $\varsigma$.
- The resulting distribution of boundary cross times can be described as the **Wiener first-passage time (WFPT) distribution**:

$$\mathcal{T} \sim \textit{Wiener}(\alpha, \beta, \tau, \delta).$$

# Current Challenges of Modeling: trial level estimation

**Issues**

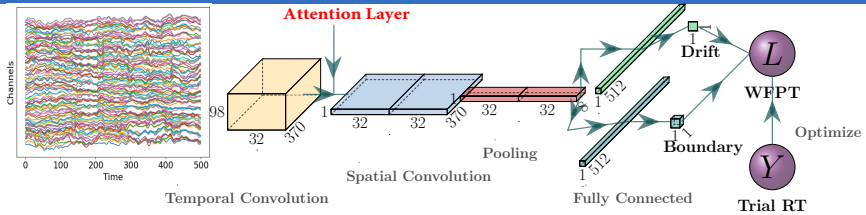
- Each trial has different speed of information processing (drift $\delta$), different level of caution (boundary $\alpha$), and different visual processing and motor execution time (non-decision time $\tau$).
- Unsolvable problem with trial estimate: more than one unknown but only have one Response Time observed.

**Goals**

- Can we introduce **single trial brain signals** as an additional known source to constrain **single trial cognitive parameters** and find out what most likely give rise to the choice response time? **Power of Inference**
- Can we construct a single neurocognitive model that can **automatically learn** to extract meaningful trial-level brain signals to predict cognitive parameters without feature engineering? **Insight of the relationship between brain and cognitive processes**

# Decision SincNet: Bridging Neural Network and Drift-Diffusion Model

# Decision SincNet Model



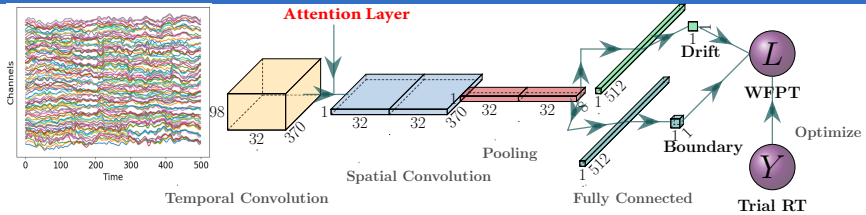

- The model takes trial EEG without prior knowledge of features and make trial predictions of decision criterion (boundary) and evidence accumulation rate (drift)
- Multi-Task Learning allows for fitting the parameters simultaneously
- With **WFPT likelihood** as a loss function, we can use gradient descent to learn an end-to-end model. Let $\mathcal{X} := \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$ and $\mathcal{T} := \{t_1, \ldots, t_N\}$ as the sets of EEG inputs and corresponding RT, we have:

$$p(\mathcal{T} \mid \mathcal{X}, \boldsymbol{\theta}) = \prod_{n=1}^{N} \text{Wiener}\left(t_n \mid \delta(\boldsymbol{x}_n, \boldsymbol{\theta}), \alpha(\boldsymbol{x}_n, \boldsymbol{\theta}), \beta, \tau\right) \tag{1}$$

The optimization can thus be expressed as $\boldsymbol{\theta}^\star = \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\theta)$ with

$$\mathcal{L}(\theta) = -\sum_{n=1}^{N} \log\left(t_n \mid \delta(\boldsymbol{x}_n, \boldsymbol{\theta}), \alpha(\boldsymbol{x}_n, \boldsymbol{\theta}), \beta, \tau\right) - \text{corr}(\boldsymbol{\delta}, \boldsymbol{t_n}).$$

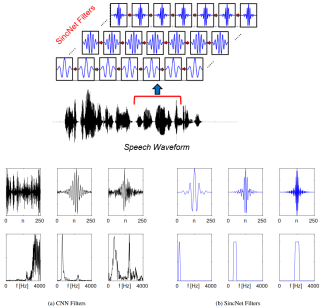
# Model Highlights: Interpretability




- SincNet parameterized the kernel as a Sinc function with only two parameters (the low and high cutoff frequencies $f_1, f_2$) to convolve with the time series samples as bandpass fitlers (Ravanelli and Bengio (2018))

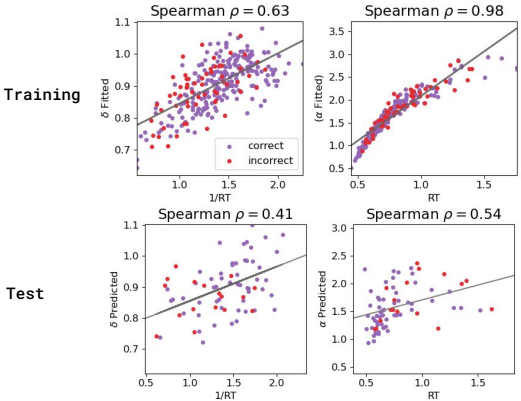
$$\boldsymbol{k}_j[n] = 2f_2 sinc(2\pi f_{2,j} n) - 2f_1 sinc(2\pi f_{1,j} n) \qquad (2)$$

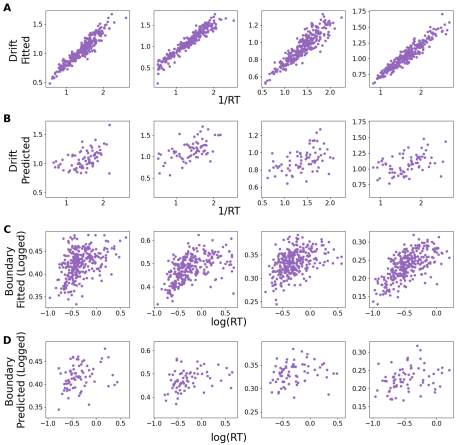
- Self-attention mechanism: a "Squeeze-and-Excitations" computational unit that uses global information to recalibrate sinc filters and produce a set of weights that rank the importance of filters (Hu et al. (2018))



# Model Evaluation and Results

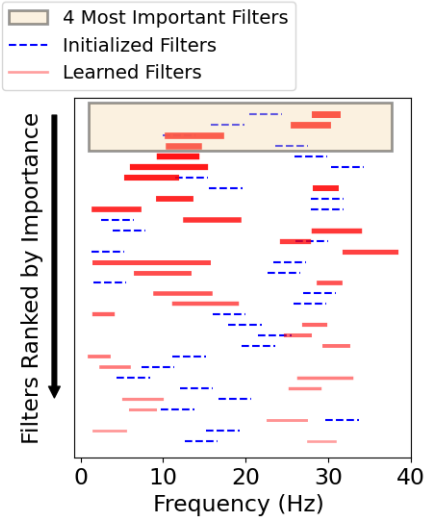
# Model Evaluation

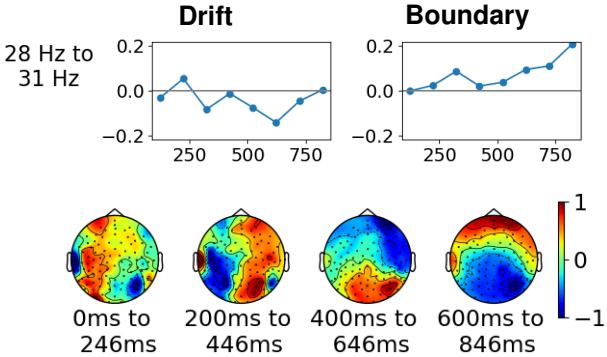



- Trial-to-trial variability: Relationships with RT
- Model Comparison: Stronger model than using behavioral data alone
- Generalization: Meaningful out-of-sample predictions

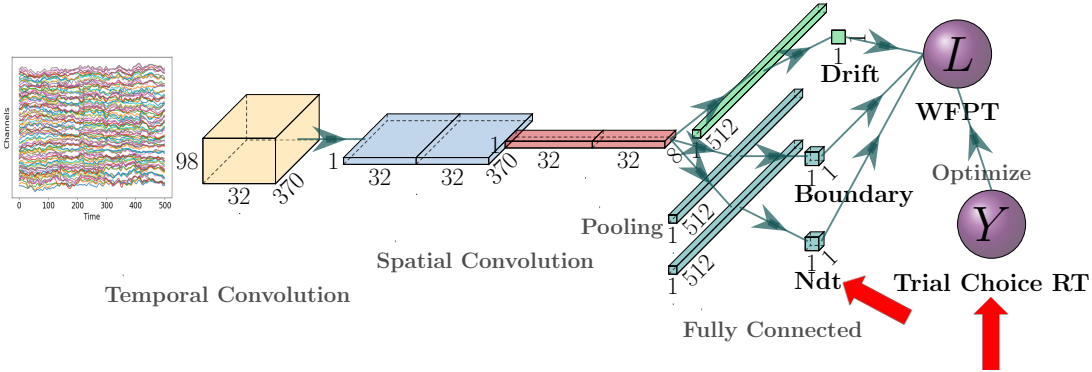
# Interpretability of Decision SincNet




- Attention weights can rank the importance of bandpass filters
- Activation map folloinwg Temporal Layer shows spatial patterns corresponding the each frequency band
- Activation map after FC Layers can reconstuct how filtered signals are weighted for making predictions.

# Recent Progress: Fitting all Parameters

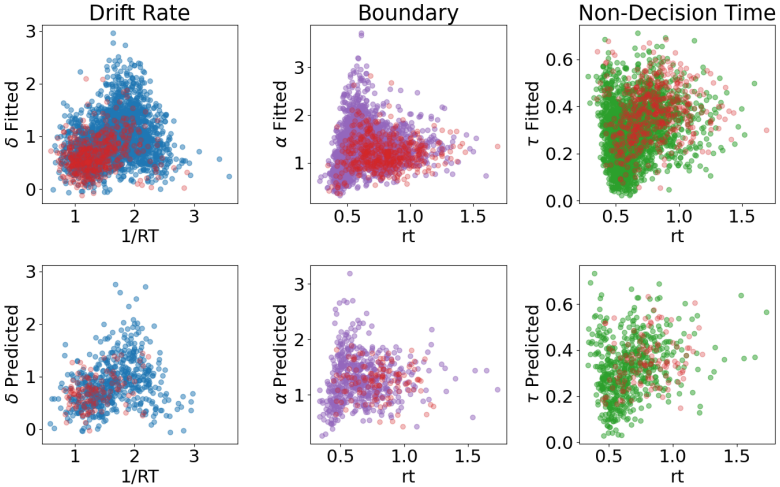



- Fitting all Drift-Diffusion Model Parameters
- Bivariate Wiener Likelihood as a loss function

# Recent Progress: Fitting all Parameters

**Training**

**Test**



# Thank you

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Corresponding Github Repo:
https://github.com/jennyqsun/SincNet-on-EEG-to-Predict-Respon
seTime

Hu, J., Shen, L., and Sun, G. (2018). Squeeze-and-excitation networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 7132–7141.

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